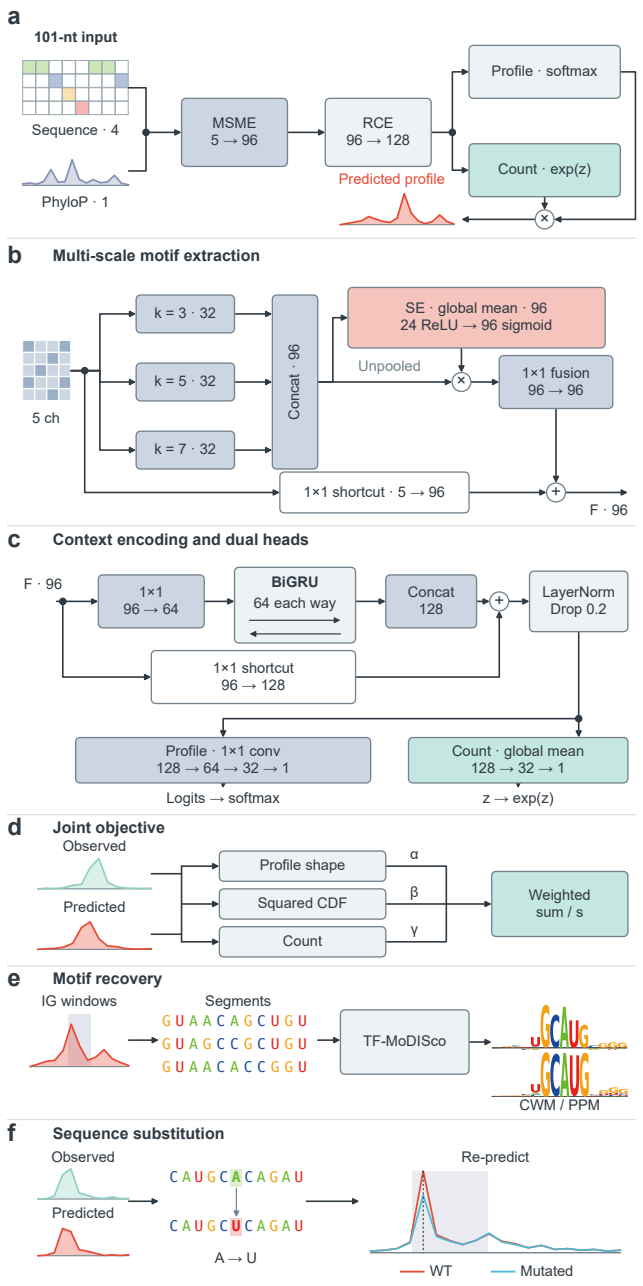

a
101-nt input
Sequence · 4
PhyloP · 1
MSME
5 → 96
RCE
96 → 128
Profile · softmax
Count · exp(z)
Predicted profile
b
Multi-scale motif extraction
5 ch
k = 3 · 32
k = 5 · 32
k = 7 · 32
Concat · 96
SE · global mean · 96
24 ReLU → 96 sigmoid
Unpooled
1×1 fusion
96 → 96
1×1 shortcut · 5 → 96
F · 96
c
Context encoding and dual heads
F · 96
1×1
96 → 64
BiGRU
64 each way
Concat
128
LayerNorm
Drop 0.2
1×1 shortcut
96 → 128
Profile · 1×1 conv
128 → 64 → 32 → 1
Logits → softmax
Count · global mean
128 → 32 → 1
z → exp(z)
d
Joint objective
Observed
Predicted
Profile shape
Squared CDF
Count
α
β
γ
Weighted
sum / s
e
Motif recovery
IG windows
Segments
G U A A C A G C U G U
G U A G C C G C U G U
G U A A C A C C G G U
TF-MoDISco
CWM / PPM
f
Sequence substitution
Observed
Predicted
C A U G C A C A G A U
C A U G C U C A G A U
A → U
Re-predict
WT
Mutated